

Survey of Reticuloendotheliosis Virus in Wild Turkeys (*Meleagris gallopavo*) in Texas, USA

Brittany Stewart,¹ Camille Trautman,¹ Faith Cox,¹ Heidi Spann,¹ Jason Hardin,² Robert Dittmar,³ and Dustin Edwards^{1,4}
¹Tarleton State University, Department of Biological Sciences, Box T-0100, Stephenville, Texas 76402, USA; ²Texas Parks and Wildlife Department, 2706 W. Commerce, Buffalo, Texas 75831, USA; ³Texas Parks and Wildlife Department, 309 Sidney Baker South, Kerrville, Texas 78028, USA; ⁴Corresponding author (email: dcedwards@tarleton.edu)

ABSTRACT: Reticuloendotheliosis virus (REV) is an immunosuppressive and sometimes oncogenic avian retrovirus that establishes lifelong infection in a wide range of avian species. REV-infected wild birds roaming near at-risk captive flocks, such as is the case for the highly endangered Attwater's prairie chicken (*Tympanuchus cupido attwateri*; APC), could act as a reservoir for viral transmission. In wild birds, prevalence rates of REV are low and appearance of associated disease is uncommon. During 2016–17, nearly half of all captive adult APC mortality at Fossil Rim Wildlife Center captive breeding facility in Glen Rose, Texas, USA was attributed to REV infection. The unusually high REV prevalence rate prompted us to survey for this virus in wild galliforms throughout the region. From 2016–17, 393 blood samples collected from two subspecies of wild turkeys (*Meleagris gallopavo*) were tested for REV proviral DNA through amplification of the viral 3' long terminal repeat and segments of the viral *pol* gene. In REV-affected counties, 5% (5/98) of native Rio Grande wild turkeys (*Meleagris gallopavo intermedia*) were identified as REV-positive. In addition, we detected REV in one of 62 eastern wild turkeys (*Meleagris gallopavo silvestris*) that had been imported during conservation efforts. To better determine protective measures, continued surveillance, including collection and genetic analysis of REV-infected samples, is necessary to identify sources of REV outbreaks in captive APC flocks.

Key words: Attwater's prairie chickens, avian retrovirus, dried blood spots, reticuloendotheliosis virus, wild turkeys.

Reticuloendotheliosis virus (REV) is an immunosuppressive and sometimes oncogenic avian retrovirus that establishes lifelong infection in Galliformes, Anseriformes, and Passeriformes (Ferro et al. 2017). The virus infects B-lymphocytes, and infected birds can exhibit clinical abnormalities such as anemia, neoplasia, nonneoplastic runting, lymphoma, feathering abnormality, and heightened

susceptibility to coinfections. However, infection is rarely associated with clinical disease in wild birds (Nair et al. 2013). The prevalence of REV in the wild has been shown to be generally low, but infected wild birds, such as wild turkeys (*Meleagris gallopavo*), roaming near captive locations could act as reservoirs for viral transmission (Peterson et al. 2002; Ingram et al. 2015; Ferro et al. 2017). Attwater's prairie chickens (*Tympanuchus cupido attwateri*; APC) at the Fossil Rim Wildlife Center (FRWC) captive breeding facility in Glen Rose, Texas, USA are tested by quantitative PCR for REV semiannually before setting breeding pairs in January and after breeding season in June. During 2016–17, nearly half of all adult APC mortality was attributed to REV infections at this facility, and most infected individuals were euthanized before symptoms appeared. The unusually high REV incidence within the endangered APC population prompted us to survey the region for evidence of this virus in related wild turkeys.

Both Rio Grande wild turkeys (*Meleagris gallopavo intermedia*) and eastern wild turkeys (*Meleagris gallopavo silvestris*) have been found to be infected by REV (Ley et al. 1989; Hayes et al. 1992; Peterson et al. 2002). Eastern wild turkeys infected with REV have been observed in close contact with domestic poultry, the probable source of infection to wild birds because domestic flocks have occasionally been administered REV-contaminated fowlpox and Marek's disease vaccines (Hayes et al. 1992; Nair et al. 2013). Wild turkeys were nearly extirpated from Texas by the early 1900s as a result of overhunting, habitat degradation, and disease (Rocke and Yuill 1987; Peterson et al. 2002). Restoration

efforts have involved translocation of turkeys from areas of higher concentration to other portions of the USA (Van Why et al. 2001). The “Super-Stocking” plan of translocating 240 individuals to restore eastern wild turkey populations in Texas is an example of this effort (Alldredge et al. 2014). Translocated birds are tested for *Mycoplasma gallisepticum*, *Mycoplasma synoviae*, *Mycoplasma meleagridis*, *Salmonella pullorum*, *Salmonella typhimurium*, Newcastle disease virus, and avian influenza virus prior to transport, but testing for REV is not standard (Charlton 2000). A study conducted in 2002 identified two REV-infected Rio Grande wild turkeys within Kerr County in central Texas; however, there have been no published reports of testing for REV in eastern wild turkeys imported into the state (Peterson et al. 2002). Our purpose was to analyze the current occurrence and location of REV in both Rio Grande and imported eastern wild turkeys in Texas.

Samples were collected from 15 January 2016 to 10 March 2016 and again from 25 January 2017 to 1 April 2017. Wild turkeys were captured using a combination of walk-in funnel traps, drop nets, and rocket nets by state wildlife agency staff authorized by an Employee Scientific Collection Authorization Permit as part of the Texas Parks and Wildlife Department eastern wild turkey restoration program and statewide banding efforts to assess harvest rates in Texas. Texas Parks and Wildlife Department biologists in 17 Texas counties collected blood on sampling cards constructed from Whatman grade 3MM paper (GE Healthcare, Chicago, Illinois, USA) prelabeled with a unique identification number. The wing-vein was punctured using a sterile lancet and blood was spotted onto a blood collection card and allowed to dry for 1 hr. Wild turkeys involved in this study were released alive and unharmed. Additional dried blood spot samples were prepared from APCs originating at FRWC and Abilene Zoological Gardens (Abilene, Texas, USA) previously tested by Texas A&M Veterinary Medical Diagnostic Laboratory (College Station, Texas, USA) and used as positive (APC B14125) and negative (APC 2909) controls for the

presence of REV proviral DNA. Five-millimeter squares were cut from each dried blood spot and washed with 1 mL of phosphate-buffered saline with 0.1% Tween 20 (PBS-T; Bio-Rad, Hercules, California, USA) in a 1.7 mL tube while rocking. The PBS-T solution was aspirated and 75 μ L of HotSHOT alkaline lysis reagent was added for genomic DNA extraction (Truett et al. 2000). The samples were incubated at 95 C for 30 min and then cooled on ice for 5 min before addition of 75 μ L of HotSHOT neutralization solution. Samples were centrifuged for 10 s and the supernatant from each was transferred to a new microcentrifuge tube. Samples were assembled for PCR in 25 μ L reactions, using 5.5 μ L nuclease-free water, 12.5 μ L OneTaq Hot Start 2X Master Mix (New England Biolabs, Ipswich, Massachusetts, USA), 1 μ L each of forward and reverse primers (10 μ M starting concentration), and 5 μ L of the eluted DNA sample (see Supplementary Materials). We used REV-specific forward and reverse primers that targeted the virus 3' long terminal repeat (LTR) to amplify proviral DNA by PCR (Aly et al. 1993). Pan-avian *GAPDH* DNA was amplified as a genomic DNA extraction control (Olias et al. 2014). For samples with detectable REV 3' LTR PCR products, we amplified REV *pol* gene segments 2500–3075 and 4777–5575 by a touchdown PCR cycle (Barbosa et al. 2007). All PCR products were resolved using gel electrophoresis in a 2% agarose gel with Tris Borate EDTA buffer (see Supplementary Materials). The PCR products of REV *pol* gene segments from identified REV-positive samples were sequenced, using the same forward and reverse primers, at Texas A&M University Corpus Christi Genomics Core Laboratory (Corpus Christi, Texas, USA). Similarity to published REV proviral sequences from APC (GenBank DQ387450) and turkey (GenBank KJ756349) samples was determined by BLASTn and sequences were aligned for analysis of nucleotide variation using MEGA X software (Altschul et al. 1990; Kumar et al. 2018).

A total of 331 Rio Grande wild turkeys were tested: 155 samples were from 15 Texas

TABLE 1. Results of reticuloendotheliosis virus by Texas county and year of reticuloendotheliosis virus proviral DNA PCR testing of dried blood spot samples collected from Rio Grande wild turkeys (*Meleagris gallopavo intermedia*; $n=331$). Percent positive and frequency was determined by the number of REV-positive samples divided by the total number of samples.

County	Year	Reticuloendotheliosis virus	
		Percent positive	Frequency
Bell	2016	0	0/12
Blanco	2016	0	0/3
Brooks	2016	0	0/2
Comal	2016	0	0/2
Cottle	2016	0	0/33
Gillespie	2016	0	0/11
Hodges	2016	0	0/4
Jones	2016	0	0/2
Kendall	2016	0	0/3
Lampasas	2016	0	0/16
Llano	2016	0	0/22
Mason	2016	3	1/30
San Saba	2016	0	0/1
Scurry	2016	0	0/5
Williamson	2016	0	0/3
Unknown	2016	0	0/6
Bell	2017	0	0/8
Blanco	2017	0	0/24
Childress	2017	0	0/4
Cottle	2017	13	1/8
Gillespie	2017	4	2/48
Jones	2017	0	0/15
Kent	2017	0	0/5
Kerr	2017	0	0/20
Llano	2017	0	0/29
Mason	2017	8	1/12
Menard	2017	0	0/2
Scurry	2017	0	0/1

counties in 2016 and 176 samples were from 12 Texas counties in 2017 (Table 1). In affected counties, REV was detected in 5% (5/98) of wild turkeys. One REV-positive bird (TSU 0208) was sampled in 2016 in Mason County, and four additional REV-positive individuals were identified in 2017 from Cottle (TSU 0079), Gillespie (TSU 0580 and 1129), and Mason (TSU 1089) counties. Fifty eastern wild turkeys were surveyed during 2016, with 45 imported from Iowa, USA and

five from West Virginia, USA. In the 2016 cohort, one REV-positive bird (TSU 0015) was detected and originated from West Virginia. The 2017 cohort included 12 eastern wild turkeys (10 imported from Missouri, USA and two imported from West Virginia, USA), none of which tested positive for REV. All imported birds appeared in good health and tested negative for avian influenza virus and Pullorum-Typhoid at the Iowa State University Veterinary Diagnostic Laboratory (Ames, Iowa, USA) or Texas A&M Veterinary Medical Diagnostic Laboratory prior to transport by air to Dallas-Fort Worth International Airport. Detection of REV proviral sequence in positive samples was further confirmed by sequencing and analysis of PCR-amplified REV *pol* gene segments (GenBank MK033877–MK033890). The REV sequences from the eastern wild turkey sample had nucleotide variations, as compared to reference sequences, at genome nucleotides 2762G>A, 2728G>A, 4924C>T, and 5198G>A.

Our purpose was to further understand the prevalence of REV in wild turkeys in Texas, both native and imported, and their potential role in REV infections in at-risk captive populations. Although REV is thought to be a low-level health threat to wild birds, infected birds can be a source of infection for the highly endangered Attwater's prairie chicken. A recent study, conducted near the Attwater Prairie Chicken National Wildlife Refuge, which is bordered by Gulf Prairies and Marshes and Post Oak Savannah ecoregions in south Texas, showed REV to be at low prevalence in 32 species of birds within close proximity of a free-range APC flock (Ferro et al. 2017). In this study, none of the wild turkeys from these regions tested positive for REV. Our study detected four infected Rio Grande wild turkeys in the Edwards Plateau and Llano Uplift ecoregions in central Texas, counties contiguous to an area previously determined to contain REV-infected wild turkeys (Peterson et al. 2002). This is also the first report of a REV-infected Rio Grande wild turkey in the Rolling Plains ecoregion in north Texas, which is near captive APCs at the

Abilene Zoological Gardens. Conservation efforts would benefit from continued surveillance for REV-positive wild birds, both resident and migratory, within reachable proximity to APCs, and further genetic analysis to determine if the viral strains found in wild birds match those in captive flocks. We also report the first case of a REV-positive eastern wild turkey that was imported into the Piney Woods ecoregion of Texas for conservation purposes. Sequence alignments of the well-conserved REV *pol* gene demonstrated a nucleotide difference in the sample that originated from West Virginia compared to those in Texas and could be used to determine the origin of new infections within the region. Although REV is currently infrequently detected in the wild, areas in which potentially infected eastern wild turkeys might be imported could be at risk for introduction of REV or increased transmission. Additional studies are necessary to determine whether imported birds should be REV-tested prior to transport. Future regional surveys of wild turkeys for other emerging retroviruses, including lymphoproliferative disease virus, which has similar pathology that is nearly indistinguishable from REV, would be valuable in APC and eastern wild turkey conservation efforts (Allison et al. 2014).

We thank the Texas Parks and Wildlife Department biologists who collected the wild turkey blood samples, Fossil Rim Wildlife Center, Abilene Zoological Gardens, and US Fish and Wildlife Service for Attwater's prairie chicken blood samples, and P. Sudman, H. Haefele, J. Swenson, and P. Ferro for their technical guidance. This work was a collaboration between Tarleton State University and Texas Parks and Wildlife Department with support from Tarleton State University Undergraduate Research Assistantships and Student Research Grants.

SUPPLEMENTARY MATERIAL

Supplementary material for this article is online at <http://dx.doi.org/10.7589/2018-08-187>.

LITERATURE CITED

- Allredge BE, Hardin JB, Whiteside J, Isabelle JL, Parsons S, Conway WC, Cathey JC. 2014. Eastern wild turkeys in Texas: Biology and management. *Texas A&M AgriLife Extension WF-011*. Texas A&M University, College Station, Texas, 20 pp.
- Allison AB, Keel MK, Philips JE, Cartoceti AN, Munk BA, Nemeth NM, Welsh TI, Thomas JM, Crum JM, Lichtenwalner AB, et al. 2014. Avian oncogenesis induced by lymphoproliferative disease virus: A neglected or emerging retroviral pathogen? *Virology* 450–451:2–12.
- Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ. 1990. Basic local alignment search tool. *J Mol Biol* 215:403–410.
- Aly MM, Smith EJ, Fadly AM. 1993. Detection of reticuloendotheliosis virus infection using the polymerase chain reaction. *Avian Pathol* 22:543–554.
- Barbosa T, Zavala G, Cheng S, Villegas P. 2007. Full genome sequence and some biological properties of reticuloendotheliosis virus strain APC-566 isolated from endangered Attwater's prairie chickens. *Virus Res* 124:68–77.
- Charlton KG. 2000. Antibodies to selected disease agents in translocated wild turkeys in California. *J Wildl Dis* 36:161–164.
- Ferro PJ, Morrow ME, Flanagan JP, Ortego B, Chester RE, Mueller JM, Lupiani B. 2017. Wild birds, a source of reticuloendotheliosis virus infection for the endangered Attwater's prairie-chicken (*Tympanuchus cupido attwateri*)? *J Wildl Dis* 53:586–590.
- Hayes LE, Langheinrich KA, Witter RL. 1992. Reticuloendotheliosis in a wild turkey (*Meleagris gallopavo*) from coastal Georgia. *J Wildl Dis* 28:154–158.
- Ingram DR, Miller DL, Baldwin CA, Turco J, Lockhart JM. 2015. Serologic survey of wild turkeys (*Meleagris gallopavo*) and evidence of exposure to avian encephalomyelitis virus in Georgia and Florida, USA. *J Wildl Dis* 51:374–379.
- Kumar S, Stecher G, Li M, Knyaz C, Tamura K. 2018. MEGA X: Molecular evolutionary genetics analysis across computing platforms. *Mol Biol Evol* 35:1547–1549.
- Ley DH, Ficken MD, Cobb DT, Witter RL. 1989. Histomoniasis and reticuloendotheliosis in a wild turkey (*Meleagris gallopavo*) in North Carolina. *J Wildl Dis* 25:262–265.
- Nair V, Zavala G, Fadly AM. 2013. Reticuloendotheliosis. In: *Diseases of poultry*, 13th Ed. Swayne DE, Glisson JR, McDougald LR, Nolan JV, Suarez DL, and Nair V, editors. Wiley-Blackwell, Ames, Iowa. pp. 593–604.
- Olias P, Adam I, Meyer A, Scharff C, Gruber AD. 2014. Reference genes for quantitative gene expression studies in multiple avian species. *PLoS One* 9: e99678.
- Peterson MJ, Aguirre R, Ferro PJ, Jones DA, Lawyer TA, Peterson MN, Silvy NJ. 2002. Infectious disease

- survey of Rio Grande wild turkeys in the Edwards Plateau of Texas. *J Wildl Dis* 38:826–833.
- Rocke TE, Yuill TM. 1987. Microbial infections in a declining wild turkey population in Texas. *J Wildl Manag* 51:778–782.
- Truett GE, Heeger P, Mynatt RL, Truett AA, Walker JA, Warman ML. 2000. Preparation of PCR-quality mouse genomic DNA with hot sodium hydroxide and tris (HotSHOT). *Biotechniques* 29:52–54.
- Van Why KR, Applegate RD, Cable TT, Gipson PS. 2001. Survey of Kansas wild turkey hunters: Experiences, opinions, and satisfactions. Kansas State University, Agriculture Experiment Station and Cooperative Extension Service, Manhattan, Kansas. 27 pp.

Submitted for publication 1 August 2018.

Accepted 29 October 2018.